

S3 Supplementary QC and Robustness

Quality control, ten-part confound audit, and reviewer-facing robustness summary

Article type: Supplement

Short title: S3 QC/Robustness — MNJ Task Geometry (ds004511)

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S3.0 Sample and epoch coverage overview

Of 45 subjects, 44 contributed valid GG data. One subject (sub-S200203) had zero valid GG epochs and was excluded from GG analyses only (no GG EEG acquisition exists for this subject; dataset README documents a recording-time technical error). This subject contributed valid CC and Rest data, so 45 subjects contributed CC data and 45 contributed Rest data. The intersection sample used for the confound audit (S3.3) requires valid Jacobian data in all three tasks and comprises $N = 44$ subjects.

Task	N subj. (valid)	Mean ep./subj.	Median ep./subj.	Range
Gambling	44 / 45	891	910	0–1164
Cognitive control	45 / 45	262	261	227–332
Rest	45 / 45	126	124	121–150

Figure 1: Epoch coverage by task ($N = 45$ subjects total). One gambling session (sub-S200203) has zero valid epochs, due to a genuinely missing GG EEG acquisition (see S1.0, S3.0), and is excluded from GG-specific analyses; its row is included in the mean/median/range above as 0 to reflect true coverage across all 45 subjects. Gambling’s much longer task duration is the motivation for the epoch-count-matched sensitivity analysis (S3.3, part B). Source: 00_sanity_check/epoch_coverage.csv.

S3.1 Physiological QC pass rates

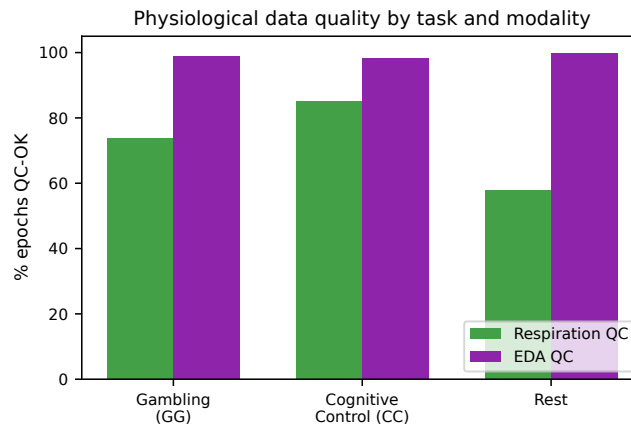


Figure 2: Respiration and EDA epoch-level QC pass rates by task (duplicated from the main text Fig. 6 for supplement completeness).

Task	Resp. QC pass rate	EDA QC pass rate
Gambling	73.4%	98.8%
Cognitive control	85.1%	98.3%
Rest	57.7%	99.7%

Figure 3: Mean epoch-level QC pass rate by task and modality, averaged across subjects. Respiration QC is substantially lower during Rest (57.7%), consistent with irregular/shallow breathing during eyes-closed passive rest; despite this, qc_ok_for_eap (the composite subject-level gate requiring adequate respiration data in at least one usable task pairing) is satisfied for 97.8% of subjects (44/45). EDA QC is uniformly high (>98%) across all three tasks because the session-level rejection gate (S1.3) removes the small number of fully bad sessions rather than sporadic epochs. Source: 00_sanity_check/resp_qc_by_subject_task.csv, subject_qc_flags.csv, 06_eda_extraction/eda_qc_summary.csv.

Five EDA sessions across three subjects were rejected outright at the session-quality gate (flat/saturated signal): sub-S201222 Rest; sub-S200303 Rest; and sub-S201210 Rest, CC, and GG. Separately, one GG session (sub-S200203) produced zero valid epochs and was excluded from GG analyses across all modalities, not only EDA (see S3.0 for the per-subject reason).

S3.2 EEG/EOG artifact balance across tasks

To rule out differential artifact contamination as an alternative explanation for the MNJ task effect (audit part F, S1.6), EEG/EOG quality flags and artifact-proxy features were tabulated by task across the intersection sample.

Metric	Gambling	Cognitive control	Rest
qc_ok_eeg (mean)	1.000	1.000	1.000
qc_ok_eog (mean)	1.000	1.000	1.000
eog_blink_rate (mean $\pm$ SD)	0.42 $\pm$ 0.30	0.44 $\pm$ 0.33	0.19 $\pm$ 0.29
eog_artifact_fraction (mean $\pm$ SD)	0.0009 $\pm$ 0.013	0.0012 $\pm$ 0.011	0.0003 $\pm$ 0.004
eeg_highfreq_power_30_45 (mean $\pm$ SD)	0.149 $\pm$ 0.647	0.144 $\pm$ 0.341	0.705 $\pm$ 3.919
eeg_hjorth_complexity (mean $\pm$ SD)	2.331 $\pm$ 0.827	2.266 $\pm$ 0.785	2.298 $\pm$ 0.562
eeg_permutation_entropy (mean $\pm$ SD)	0.905 $\pm$ 0.021	0.908 $\pm$ 0.021	0.873 $\pm$ 0.050

Table 1: EEG/EOG artifact-proxy features by task, intersection sample ($N = 44$). Both qc_ok_eeg and qc_ok_eog are at or near 1.0 for all tasks (no differential quality gate failure). Blink rate is markedly lower at Rest (eyes closed, fewer voluntary blinks) — the opposite direction from what would be needed to explain higher GG deformation via blink artifact. High-frequency (30–45 Hz, EMG-contamination proxy) power is **higher** at Rest, not GG, ruling out muscle-artifact contamination as an explanation for the GG-selective Frobenius effect. Hjorth complexity and permutation entropy are closely matched across all three tasks. Source: 05_mnj_confound_audit/artifact_balance.csv.

S3.3 Ten-part robustness and confound audit (full results)

This section reports the complete numerical results underlying the audit protocol described in S1.6, run in response to the science-lead review (sciencelead/001.md, sciencelead/002.md).

(A) Intersection sample. Restricting the primary MNJ contrasts to the $N = 44$ subjects with valid data in all three tasks:

Metric	Contrast	d	FDR q	Sig.
frobenius_norm	GG vs CC	1.620	5.6e-12	Yes
frobenius_norm	GG vs Rest	0.738	2.8e-5	Yes
frobenius_norm	CC vs Rest	-0.263	0.196	No
spectral_radius	GG vs CC	1.808	5.6e-12	Yes
rotational_power	GG vs CC	-1.025	1.1e-7	Yes
aci	GG vs CC	-1.015	1.0e-7	Yes
mdr	GG vs CC	0.420	0.006	Yes

Table 2: Selected rows from the intersection-sample re-analysis (all 21 rows in `mnj_intersection_wilcoxon.csv`); the GG vs. CC Frobenius effect is numerically identical to the full-sample result because GG’s own valid-subject set already coincides with $N = 44$. Only the CC vs. Rest contrast changes appreciably (full sample $N = 45$, $d = -0.229$, $p_{\text{fdr}} = 0.243$ vs. intersection $N = 44$, $d = -0.263$, $p_{\text{fdr}} = 0.196$) — a negligible, non-significant difference in both cases.

(B) Epoch-count matching. Subsampling each subject to the minimum epoch count across their three tasks (median ≈ 122 epochs) before recomputing medians:

Metric	Contrast	Full-sample d	Epoch-matched d	Sig. (matched)
frobenius_norm	GG vs CC	1.620	1.444	Yes
frobenius_norm	GG vs Rest	0.738	0.690	Yes
spectral_radius	GG vs CC	1.808	1.668	Yes
rotational_power	GG vs CC	-1.025	-0.685	Yes
aci	GG vs CC	-1.015	-0.687	Yes
mdr	GG vs CC	0.420	0.461	Yes

Table 3: Epoch-count-matched sensitivity analysis. All primary GG-involving contrasts remain significant after equalizing epoch counts across tasks; effect sizes shrink modestly (Frobenius: -11% ; rotational power/ACI: -33%) but no contrast changes sign or significance direction. This confirms that GG’s much larger epoch count (mean 891 vs. 262 for CC, 126 for Rest) is not the primary driver of the observed effects. Full 21-row table: `05_mnj_confound_audit/mnj_epoch_matched_wilcoxon.csv`.

(C) Leave-one-subject-out (LOOSO).

Metric	Full d	LOO min d	LOO max d	Sign-consistent
frobenius_norm	1.620	1.595	1.711	44/44
spectral_radius	1.808	1.781	1.943	44/44
rotational_power	-1.025	-1.112	-1.003	44/44
aci	-1.015	-1.096	-0.993	44/44

Table 4: LOOSO robustness for the four primary GG vs. CC effects ($N = 44$ iterations each). All four metrics retain their sign in 100% of leave-one-out iterations, and all 44 LOO iterations remain significant at $p < 0.05$ (`loo_frac_p05 = 1.0` for all four metrics). No single subject drives the effect. Source: `05_mnj_confound_audit/mnj_looso.csv`.

(D) Bootstrap confidence intervals (2000 resamples, 95% CI on Cohen’s d):

Metric	d	95% CI	Excludes 0
frobenius_norm	1.620	[1.306, 2.128]	Yes
spectral_radius	1.808	[1.451, 2.441]	Yes
rotational_power	-1.025	[-1.415, -0.743]	Yes
aci	-1.015	[-1.384, -0.745]	Yes

Table 5: Bootstrap 95% confidence intervals for the four primary metrics ($N = 44$, percentile method). All four intervals exclude zero by a wide margin. Source: `05_mnj_confound_audit/mnj_bootstrap.csv`.

(E) Within-subject permutation test (5000 sign-flip permutations):

Metric	Observed median Δ	Permutation p
frobenius_norm	+0.0277	< 0.0002
spectral_radius	+0.0156	< 0.0002
rotational_power	-0.0304	< 0.0002
aci	-0.0513	< 0.0002

Figure 4: Within-subject sign-flip permutation test (5000 permutations). 0/5000 label-shuffled permutations produced a median difference as extreme as the observed one, for any of the four primary metrics; exact $p = 0$ is reported as < 0.0002.

Source: 05_mnj_confound_audit/mnj_permutation.csv.

(F) Artifact balance: see S3.2.

(G) Physiological confound partial correlations: the initial physiological-confound set contained five candidates (HR, respiration rate, respiratory anchor index, EOG blink rate, tonic EDA); the combined seven-row table, extended with corrugator EMG (part H) and Sync-pulse event rate (part J), is reported in S2.6. None of the seven candidate confounds survives BH-FDR correction (all $q > 0.20$).

(H) Corrugator-EMG confound: see S3.4b. Corrugator EMG does not differ between GG and CC ($d = 0.08$, $p = 0.51$) and does not correlate with the GG-CC Frobenius difference ($r_s = -0.08$, $p = 0.60$).

(I) Conventional-EEG baseline comparison: see S3.4c. The largest of 16 conventional-EEG GG-vs-CC effect sizes (relative theta power, $d = 0.63$) is less than half the primary MNJ Frobenius effect ($d = 1.62$).

(J) Event/response-rate sensitivity: see S3.4d. Session-level Sync(1) pulse rate is higher in CC than GG (opposite of a naive event-density account), the GG-CC event-rate difference is uncorrelated with the GG-CC Frobenius difference ($r_s = -0.05$, $p = 0.77$), and the effect is undiminished ($d = 1.70$ vs. $d = 1.62$) when restricted to below-median-event-density epochs in both tasks.

S3.4 EDA, corrugator-EMG, conventional-EEG, and event-rate follow-ups

S3.4a EDA confound follow-up (Script 07 detail)

The EDA confound test was run as a dedicated follow-up (07_eda_confound_control.py) once the standalone EDA extraction (Script 06) was complete, extending the four-confound table from Script 05 to five confounds.

Test	r_s	p	N
EDA tonic SCL diff. $\times$ Frobenius diff. (GG-CC)	-0.10	0.51	43

Figure 5: EDA-specific confound test: Spearman correlation between the subject-level GG-CC difference in tonic EDA (SCL) and the GG-CC difference in MNJ Frobenius norm. The null result ($r_s = -0.10$, $p = 0.51$) is consistent with, and independently corroborates, the direct EDA GG-vs-CC comparison reported in the main text R5 ($d = -0.07$, $p = 0.55$): tonic sympathetic arousal is statistically indistinguishable between gambling and cognitive control both in absolute level and in its subject-level covariation with the neural flow-geometry effect. Source: 07_eda_confound_control/eda_confound_partial.csv.

S3.4b Corrugator-EMG confound follow-up (Script 10 detail)

Once the BioPac EMGA/EMGB corrugator-supercilii channels were added to the NMD ingestion config (config_ingest_ds004511.yaml, physio_tsv_inject) and the pipeline was re-run, 10_corrugator_emg_confound.py computed the direct GG/CC/Rest contrast on tonic emg_rms and its confound-partial test against the Frobenius difference, extending the five-confound table to six.

Test	d or r_s	p	N
emg_rms GG vs CC (direct contrast)	$d = 0.08$	0.51	44
emg_rms GG vs Rest (direct contrast)	$d = 0.54$	< 0.001	44
emg_rms CC vs Rest (direct contrast)	$d = 0.57$	< 0.0001	44
EMG diff. $\times$ Frobenius diff. (GG–CC)	$r_s = -0.08$	0.60	44

Figure 6: Corrugator-EMG confound test. Tonic corrugator EMG is matched between GG and CC (like EDA) but, unlike EDA, is elevated in both active tasks relative to rest — plausibly reflecting general facial/postural muscle tone during eyes-open tasks versus eyes-closed rest, rather than a task-specific affective signal. The confound-partial test ($r_s = -0.08$, $p = 0.60$) shows no relationship between the subject-level GG–CC EMG difference and the GG–CC Frobenius difference, closing the corrugator-EMG confound question raised for the primary effect. Source: 10_corrugator_emg_confound/emg_descriptive_wilcoxon.csv, emg_confound_partial.csv.

S3.4c Conventional-EEG baseline comparison (Script 09 detail)

Conventional-EEG feature (GG vs CC)	d_z	p	FDR q
Relative theta power	0.63	1.7e-4	3.5e-4
Alpha/theta ratio	-0.38	0.024	0.038
Spectral-edge frequency (95%)	-0.36	0.008	0.015
Theta/alpha ratio	0.31	0.037	0.056
Hjorth complexity	0.29	0.030	0.047

Figure 7: Largest five (of 16) conventional-EEG GG-vs-CC effect sizes, $N = 44$ intersection sample. For comparison, the primary MNJ effects on the same sample are Frobenius norm $d = 1.62$ and spectral radius $d = 1.81$ — roughly $2.6\text{--}4.5 \times$ larger than the strongest conventional-EEG metric. 31/48 conventional-EEG contrasts (all three task pairs) survive BH-FDR at 5%, confirming these are genuine, well-powered effects and not just power-law noise; they are simply much smaller than the MNJ geometric effect for the GG-vs-CC contrast specifically. Source: 09_conventional_eeg_baseline/conventional_eeg_wilcoxon.csv, conventional_vs_mnj_summary.csv.

S3.4d Event/response-rate sensitivity follow-up (Script 11 detail)

Raw BIDS events.tsv Sync(1) trigger-pulse onsets — dataset-native and independent of NMD ingest — were joined to the H5 epoch windows to test whether the two tasks’ differing event/response-rate structure could explain the GG>CC Frobenius effect (11_event_density_sensitivity.py; see main-text §R5b and S1.6(J) for method).

Test	d_z or r_s	p	N
Session event rate: GG vs CC (direct contrast)	$d_z = -4.00$	< 0.0001	44
Session event rate: GG vs Rest (direct contrast)	$d_z = 19.8$	< 0.0001	44
Session event rate: CC vs Rest (direct contrast)	$d_z = 16.3$	< 0.0001	44
Event-rate diff. $\times$ Frobenius diff. (GG-CC)	$r_s = -0.05$	0.77	44
Epoch event count $\times$ Frobenius (within GG)	$r_s = 0.03$	$2.1e-9$	39,985 epochs
Epoch event count $\times$ Frobenius (within CC)	$r_s = 0.03$	$6.8e-4$	11,363 epochs
GG vs CC Frobenius, low-density epochs only	$d_z = 1.70$	< 0.0001	44

Figure 8: Event/response-rate sensitivity results. Median session-level Sync(1) pulse rate: GG = 0.357 Hz, CC = 0.453 Hz, Rest = 0.002 Hz — CC’s per-second rate is **higher** than GG’s, the opposite of a naive “GG is event-dense” account (GG has far more total pulses per session, 1,285 vs. 470, but a proportionally much longer session, 60 vs. 17 min median). The subject-level GG-CC event-rate difference is uncorrelated with the GG-CC Frobenius difference. A small but statistically significant epoch-level coupling exists within both tasks ($r_s \approx 0.03$, $r^2 \approx 0.001$), but restricting the GG-vs-CC contrast to each subject’s below-median-event-density epochs in both tasks (median 644 GG / 162 CC epochs retained per subject) leaves the effect materially unchanged ($d = 1.70$ vs. $d = 1.62$ full-sample). Source: 11_event_density_sensitivity/session_event_rate.csv, event_rate_wilcoxon_by_task.csv, event_rate_confound_partial.csv, epoch_density_frobenius_correlation.csv, density_matched_wilcoxon.csv.

This section narrows, but does not fully close, the event/response-rate question: the confound-partial and density-matched tests are consistent with a genuine task-geometry effect rather than an event-density artifact, but the small residual epoch-level coupling and the dataset’s lack of stimulus/response pulse labeling mean this remains an open, if substantially narrowed, caveat (see main-text Limitations).

This completes the confound and baseline audit requested during science-lead review: all seven tested plausible peripheral-physiological/temporal-structure explanations for the GG>CC Frobenius effect (heart rate, respiration rate, respiratory anchoring, EOG blink rate, tonic EDA, corrugator EMG, event/response rate) have been tested; none accounts for the effect, and a matched conventional-EEG baseline comparison confirms the MNJ effect is not a simple restatement of band-power differences. The event/response-rate check narrows rather than fully eliminates this alternative (see main-text Limitations).

S3.5 Reviewer-facing robustness summary

Robustness check	Result	Interpretation
Intersection sample ($N = 44$)	$d = 1.62$, unchanged	Not driven by unequal task coverage
Epoch-count matching	$d = 1.44$ ($\approx 11\%$ shrink)	Not driven by GG's larger epoch count
LOOSO (44 iterations)	44/44 sign-consistent	Not driven by any single subject
Bootstrap 95% CI	[1.31, 2.13]	Excludes zero with wide margin
Permutation test (5000)	$p < 0.0002$	Exceeds all label-shuffled nulls
Artifact balance	No differential contamination	Not an EEG/EOG artifact
Heart rate confound	$r_s = -0.06$, $q = 0.83$	Not a cardiac-arousal confound
Respiration rate confound	$r_s = -0.29$, $q = 0.20$	Largest uncorrected signal; does not survive FDR
Respiratory anchor confound	$r_s = +0.33$, $q = 0.20$	Does not survive FDR
EOG blink rate confound	$r_s = -0.20$, $q = 0.44$	Not an ocular-artifact confound
Tonic EDA confound	$r_s = -0.10$, $q = 0.83$	Not a sympathetic-arousal confound
Corrugator EMG confound	$r_s = -0.08$, $q = 0.83$	Not a facial-muscle/affect confound
Conventional-EEG baseline	Largest $d = 0.63$ vs. MNJ $d = 1.62$	Not recoverable from simple band power/complexity
Event-rate confound	$r_s = -0.05$, $q = 0.77$	Not an event/response-density confound (narrowed, not fully closed)
Density-matched sensitivity	$d = 1.70$ (low-density epochs only)	Undiminished when event-dense epochs excluded
Block-level cross-check	$d = 1.70$, independent pathway	Replicates in independent aggregation

Table 6: Reviewer-facing summary of the ten-part robustness and confound audit, with individual confound rows expanded, for the primary GG>CC MNJ Frobenius-norm effect (report row: Frobenius norm unless noted; more than ten rows appear here because parts 7 and 10 each expand into several individual confound rows). All checks are consistent with a genuine, robust task-geometry effect rather than a sampling, artifact, peripheral-physiological, conventional-EEG-power, or event/response-rate confound.

Bibliography